Premier League Final Table Prediction Model

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Why This Project Matters

Project Overview

- Goal:
- - Predict final Premier League points and table positions using past seasons.

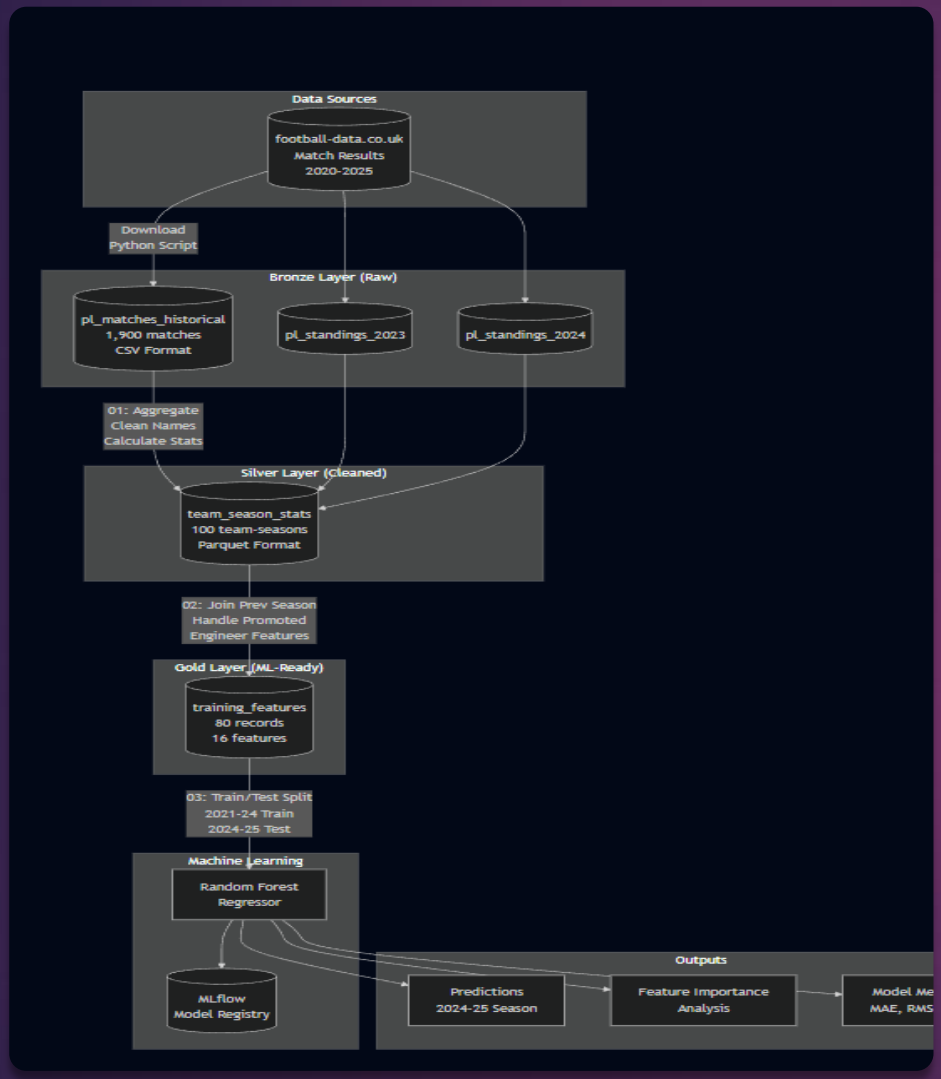
- Main questions:
- - Can a model beat a simple rule: "teams repeat last season's points"?
- - How close can we get to the real final table?
- - Which past season stats seem to matter the most?

Data Pipeline: Bronze, Silver, Gold

- Bronze:
- - Raw match level files from football-data.co.uk.
- - Combined five Premier League seasons into one table.

- Silver:
- - Cleaned column names and data types.
- - Turned match data into team by season totals.

- Gold:
- - training_features: one row per team per season with features.
- - model_predictions_2024_25: model results for the 2024–25 season.



Data Modeling pipeline

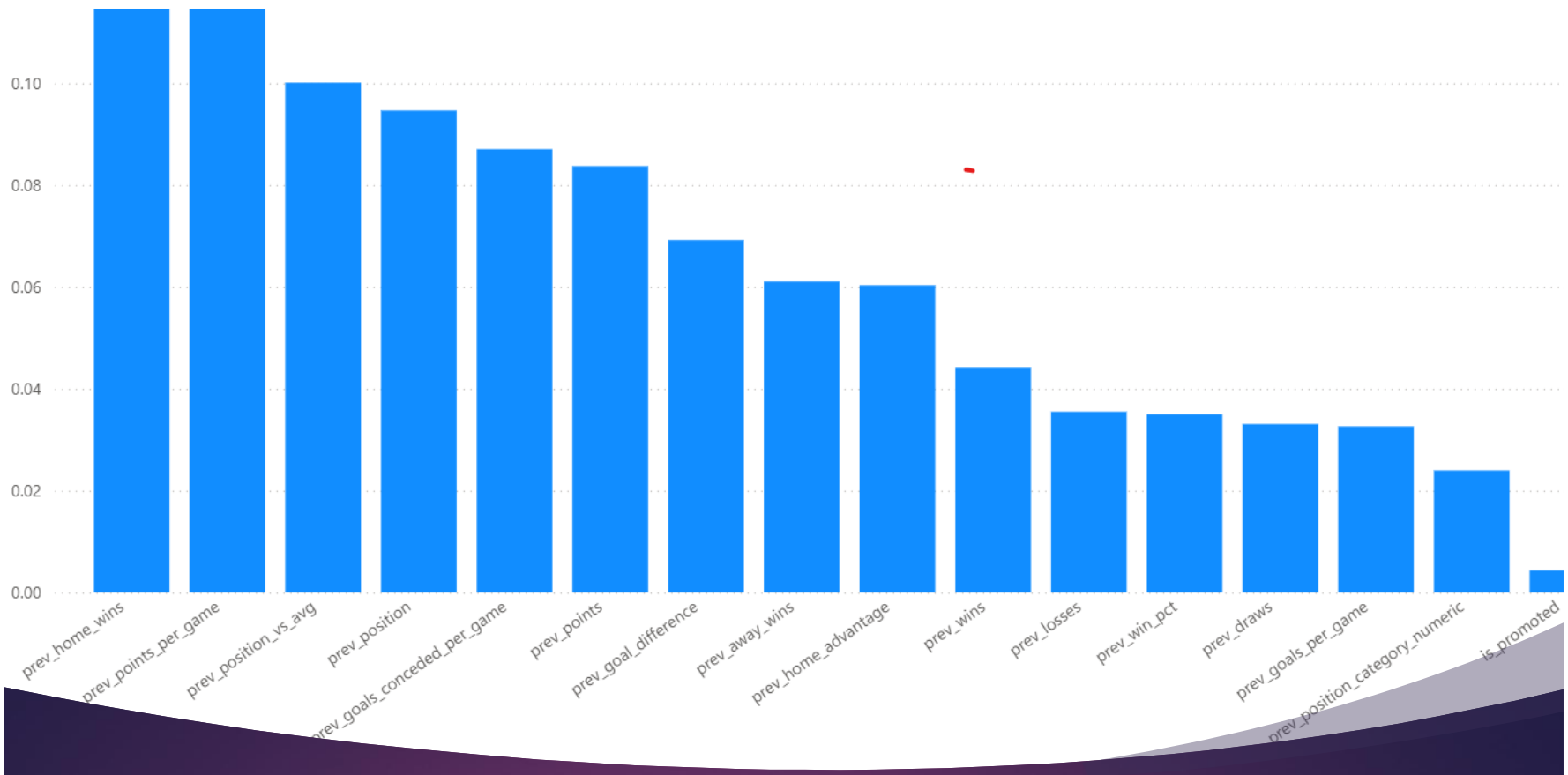
19	26	7	9	7	38	9	46
24	18	9	5	14	38	-6	46
28	19	10	6	9	38	-22	55
18	27	5	10	10	38	22	36
34	21	10	6	8	38	-25	66
20	23	4	11	8	38	-1	48
25	18	7	3	13	38	-26	53
33	34	9	10	5	38	8	54
20	34	3	11	6	38	18	50
22	39	3	10	9	38	26	42
15	40	2	14	5	38	51	32
16	35	0	12	11	38	29	44
29	20	9	6	9	38	-16	62
36	8	16	2	2	38	-43	63
43	19	11	4	7	38	-21	68
25	33	6	8	8	38	23	45
37	20	12	2	11	38	-41	76
25	30	6	9	8	38	15	47
27	15	9	5	9	38	-16	52
31	26	9	9	3	38	13	48
25	23	11	7	6	38	-2	54
35	26	9	6	7	38	-8	56
21	23	4	7	15	38	-2	44
28	16	9	2	14	38	-19	53
11	39	3	12	11	38	43	[illegible]
[illegible]	23	8	4	15	38	4	[illegible]
[illegible]	[illegible]	13	2	6	38	[illegible]	[illegible]

Raw data table

Feature Engineering

- Example input features (from previous season):
- - Points, position and position vs league average.
- - Goal difference, wins, draws, losses.
- - Points per game and win percentage.
- - Goals scored and conceded per game.
- - Home advantage, home wins and away wins.
- - Promoted team flag and position category (top 4, mid table, etc.).

- Target:
- - Final points in the next season (target_points).


0.10
0.08
0.06
0.04
0.02
0.00
prev_home_wins
prev_points_per_game
prev_position_vs_avg
prev_position
prev_goals_conceded_per_game
prev_points
prev_goal_difference
prev_away_wins
prev_home_advantage
prev_wins
prev_losses
prev_win_pct
prev_draws
prev_goals_per_game
prev_position_category_numeric

Features

Modeling Approach

- Model:
- - Random Forest Regressor to predict final points.

- Train / test setup:
- - Train on seasons 2021–22, 2022–23 and 2023–24.
- - Test on season 2024–25.

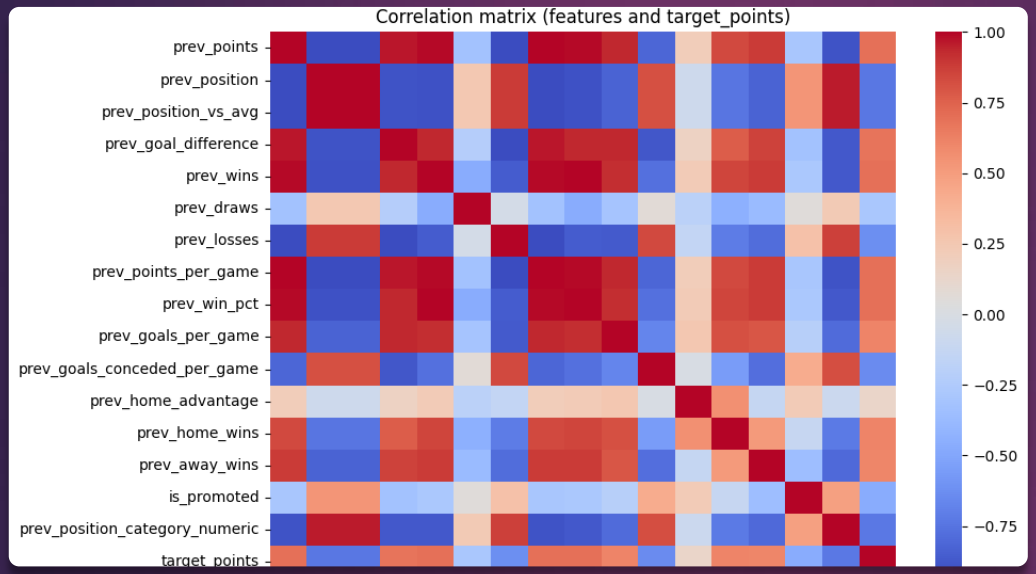
- Baseline:
- - Assume each team repeats its previous season points (prev_points).

Model Performance: Points

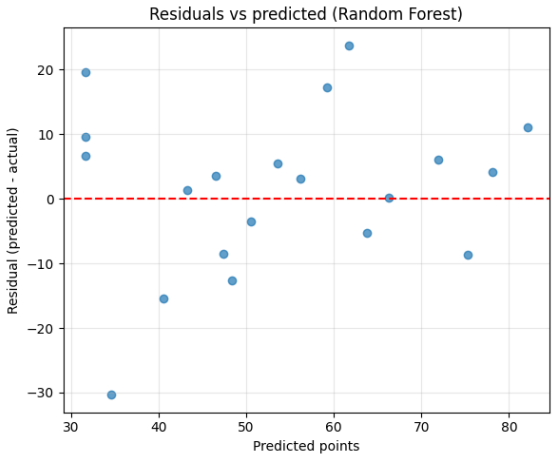
- Baseline (last season's points):
- - Mean absolute error (MAE): about 13.2 points per team.
- - R^2: about 0.18.

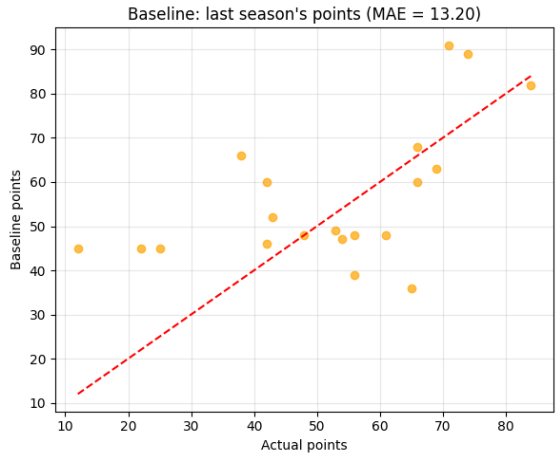
- Random Forest model:
- - Mean absolute error (MAE): about 9.8 points per team.
- - R^2: about 0.52.

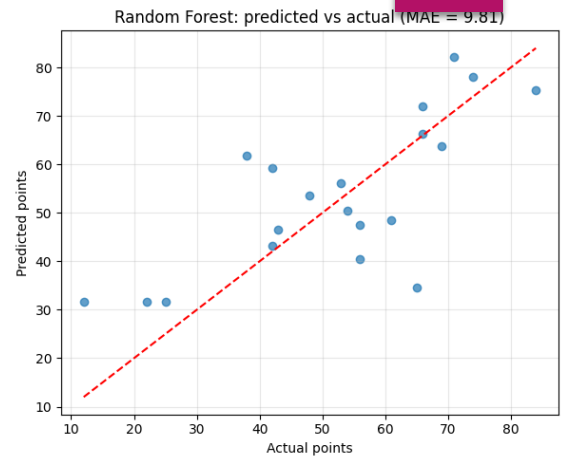
- What this means:
- - On average the model is roughly 3 to 4 points closer per team
- than the simple baseline.
- - The model explains a bit more than half of the differences in final points.



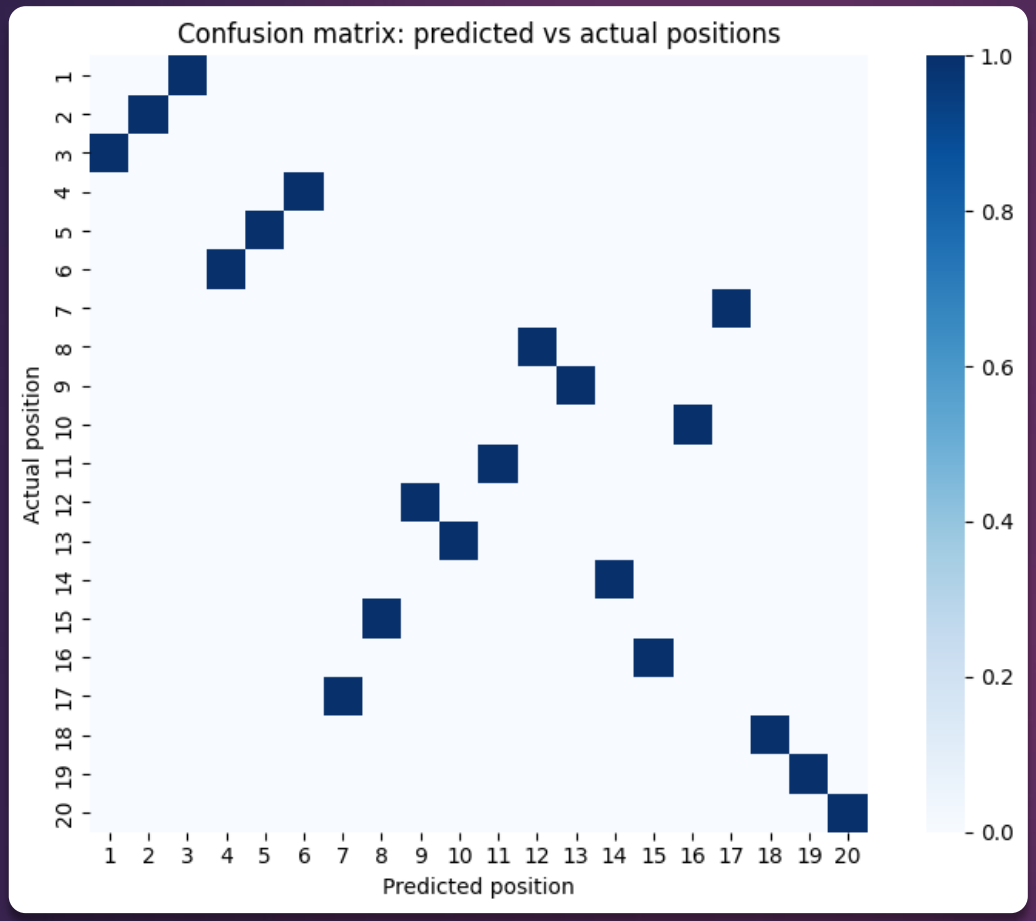
Correlation Metrix







Model performance



Confusion matrix

Model Performance: League Positions

- For the 2024–25 season:

- - Mean absolute position error: about 2.8 places.
- - Median position error: 2 places.

- Key results:
- - Correctly identified 3 of the top 4 clubs.
- - Correctly identified all 3 relegated clubs.

- So the model does a reasonable job at ranking teams,
- especially at the top and bottom of the table.

Which Features Matter Most

- Top features from the model:
- - Previous season points.
- - Previous goal difference.
- - Previous position and position vs average.
- - Points per game and win percentage.
- - Home wins and home advantage.

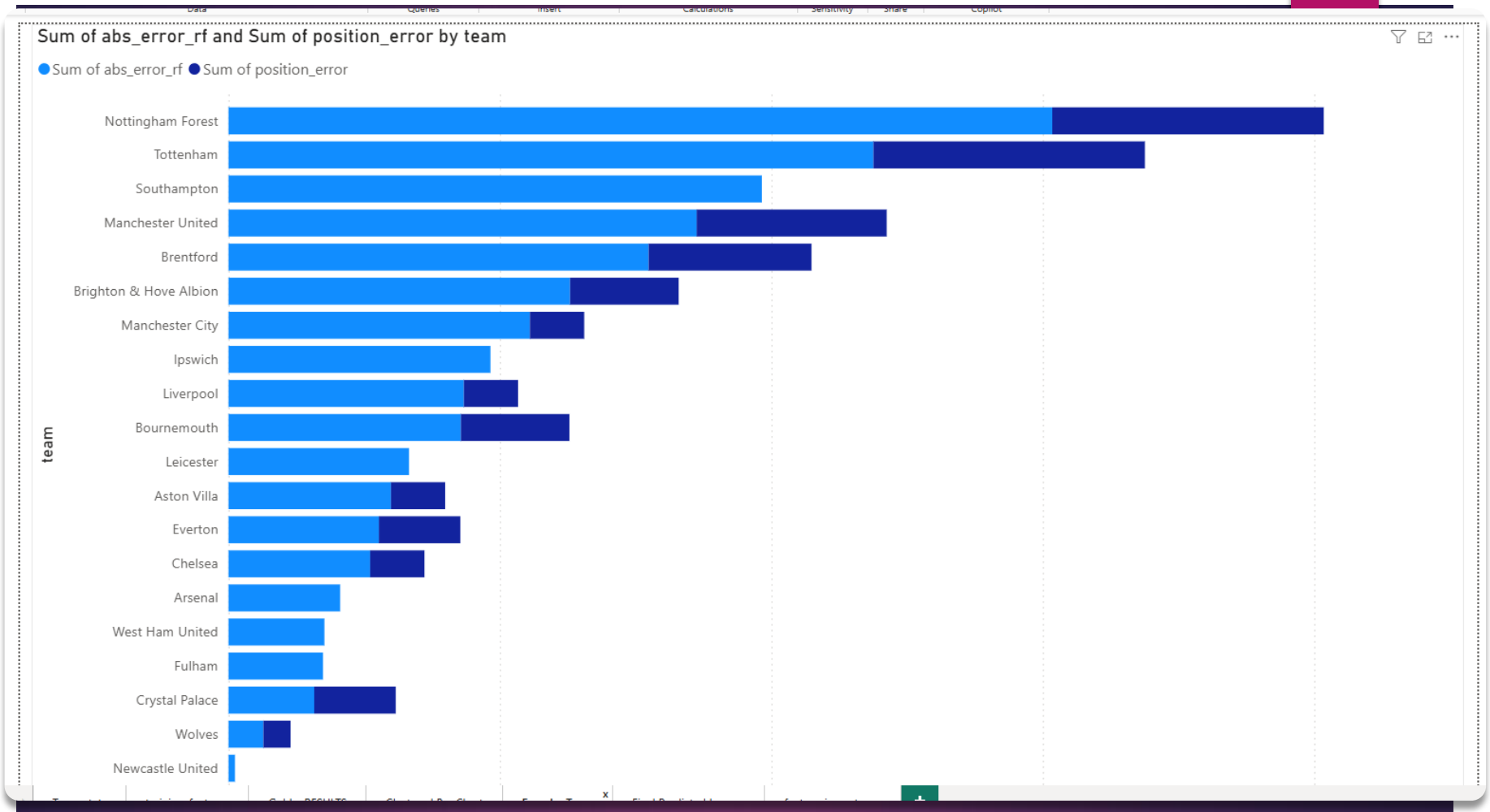
- These match football common sense:
- teams that were strong last year are likely to be strong again.

Power BI Report

- I built one Power BI page on the gold tables:

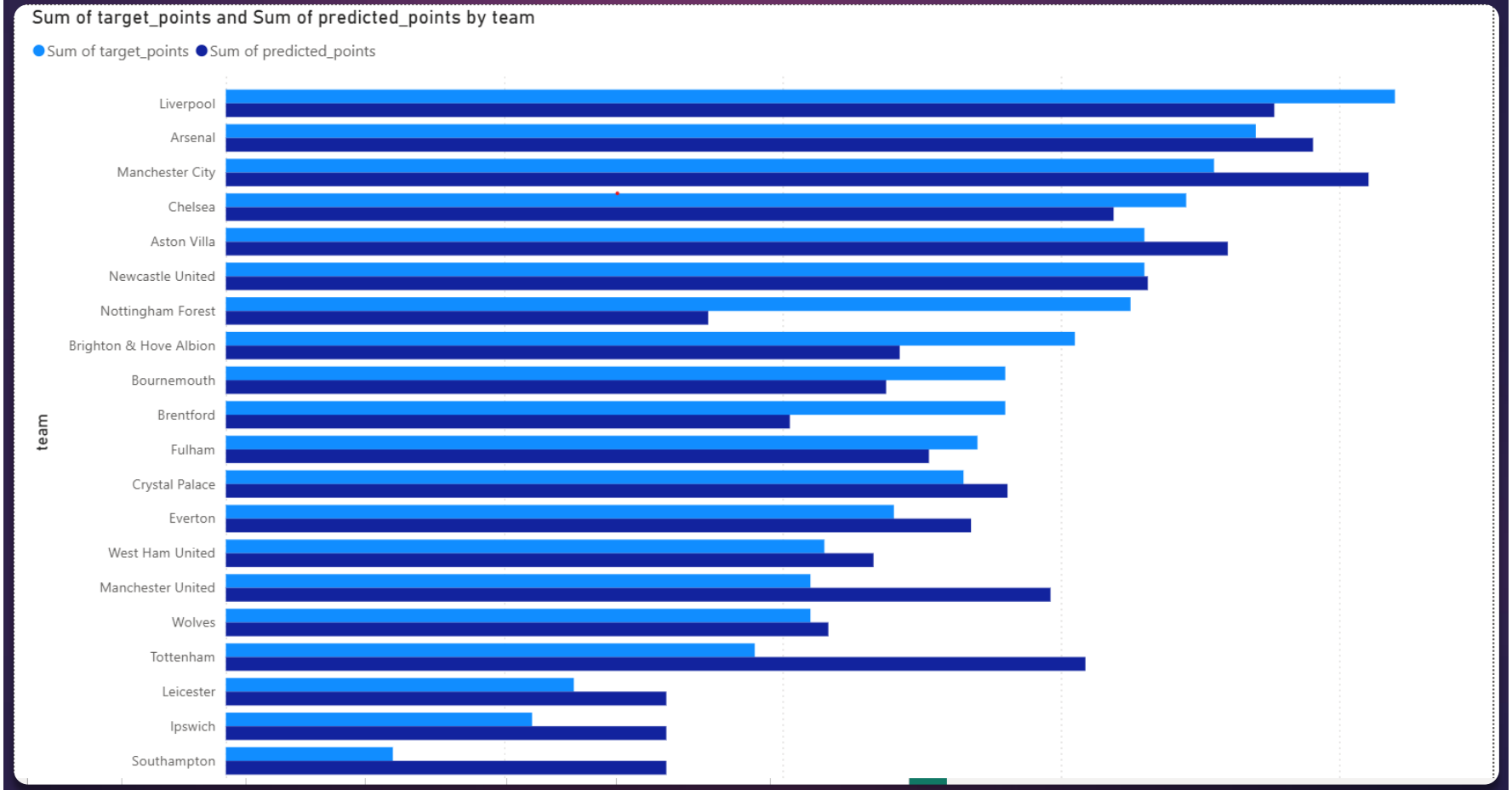
- - Scatter plot: actual points vs predicted points.
- - Clustered bar chart: actual vs predicted points by team.
- - Bar chart: absolute error by team.
- - Table: predicted vs actual league table with position error.
- - Line plot from training_features: team points across seasons.

- These visuals let me quickly show how well the model works
- and where it still misses.



Error by Team

Target point and predicted points



Limits and Next Steps

- Limits:
- - Only three past seasons used for training.
- - No match level or player level information.
- - Does not include transfers, injuries or manager changes.

- Possible next steps:
- - Add more historical seasons.
- - Bring in extra features such as xG, squad age and squad value.
- - Try other models like gradient boosting.
- - Simulate the season many times to get chances for top 4 or relegation.

Why This Project Matters

- • Football has billions of fans and is watched on every continent.
- • Stars like Cristiano Ronaldo and Lionel Messi show how big the sport is.
- • Every season, fans, analysts, and media try to predict the final table.
- • Most predictions are based on emotion, bias, or simple rules.
- • This project uses historical data and machine learning to make those
- predictions more objective and evidence-based.

Takeaways

- - Set up a small data pipeline from raw match files to clean season tables.
- - Built simple but useful features from previous seasons.
- - Trained a Random Forest model that beats a basic baseline.
- - Model is usually within 2 to 3 places of the real table.
- - Power BI makes the results easy to show in a short presentation.

- Overall, this is a first step toward more serious football prediction
- based on data instead of only opinion.